

Deconstructing and Reconstructing a Human: A Theoretical Framework

1. Physical Principles of Matter Disassembly

Quantum field theory tells us matter is composed of fermions and bosons. Disassembling a human into fundamental particles requires extreme energy. No-cloning theorem forbids copying unknown quantum states. Quantum decoherence occurs almost instantly in macroscopic systems. The Bekenstein bound suggests $\sim 10^{45}$ bits to describe a person. Landauer's principle implies huge energy costs ($\sim 10^{24}$ J).

2. Confining and Organizing Free Particles

Charged particles may be confined using electromagnetic traps (e.g. Penning traps). Scaling to 10^{28} particles is impractical due to density limits. Neutrons decay in ~ 15 minutes, neutrinos escape. Cryogenics and vacuum stability are major challenges. Radiation shielding is required against cosmic rays and solar flares.

3. Long-Term Storage Architecture

A deep-space station design would include nested vacuum chambers, superconducting magnets, and cryogenic cooling. BASE-STEP stored 70 particles in a 1-ton device. Scaling is infeasible. Quantum state preservation requires error correction beyond current capabilities. Cryogenic systems and power supply would need gigawatt reactors.

4. Reconstruction of the Human

Reassembly requires reversing disassembly via unitary operations. Energy costs for binding processes are $\sim 10^{15}$ J. Entropy reduction requires massive cooling and heat dumping. Quantum precision control of 10^{28} particles is impossible today. Quantum tomography and teleportation are unfeasible at this scale.

5. Engineering Estimates and Resources

Mass-energy equivalence: 70 kg $\rightarrow 6.3 \times 10^{18}$ J. Fragmentation energy $\sim 10^{17}$ - 10^{18} J. Landauer cost $\sim 10^{24}$ J. Scaling vacuum chambers and magnets implies trillions of dollars. Human capital: decades, thousands of scientists, global-scale funding.

6. Current Feasibility and Open Problems

Macroscopic quantum state preservation is not possible today. Quantum tomography and control scale exponentially. Energy, resources, and costs exceed all current projects. Breakthroughs in quantum error correction, cryogenics, and computation are required. In practice, this remains speculative thought experiment only.

Illustrative Costs of Related Technologies

System/Component	Scale/Specs	Cost Estimate
Large Hadron Collider (LHC)	27 km, 14 TeV	\$3×10 ⁹
Future Circular Collider	91 km, 100+ TeV	\$1.5×10 ¹⁰
DOE Frontier Supercomputer	1.1 exaFLOPS	\$600M
IBM Quantum Computer	100k qubits	\$100M
BASE-STEP Trap	1 ton, 70 particles	\$10-100M